

SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

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Summary table

Framework ID	Status	Fitness	Generation	Parent	Name
fw_b9e7d103d0	ELABORATING	0.000	0	-	Ergodic Circuit Frame- work
fw_28b4bfb95f	ELABORATING	0.000	0	-	TROPICAL_FOURIER

Details

Ergodic Circuit Framework (fw_b9e7d103d0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** COMM_COMPLEXITY
- **Target invariant:** $\text{communication_entropy_barrier}$ (real number ≥ 0) $\rightarrow$ bounds The minimum communication complexity in a k -party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove $\omega(\log n)$ lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

Primitives: - $\text{measurable_dynamical_circuit}$ (tuple (C, X, μ, T)): A Boolean circuit C where gates are labeled by operations over a probability space (X, μ) , and $T: X \rightarrow X$ is a measure-preserving transformation representing orbit_signature (function: $N \rightarrow [0,1]$): For a fixed input $x \in \{0,1\}^n$, the orbit signature is the sequence $(\mu(C(T^k(x))))_k$, capturing how the circuit's output evolves under repeated application. - $\text{cross_correlation_flow}$ (matrix $\in R^{\{d \times d\}}$): For a depth- d circuit, this matrix $F_{\{i,j\}} = \int |C_i(x) - C_j(T^{j(x)}(x))|^2 d\mu(x)$ measures how perturbations via T propagate through layers. Computable via

Tentative axioms: - A1: Any circuit with $o(n)$ communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to AC^0 circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high $\text{cross_correlation_flow}$ norm, then its communication complexity is $\Omega(n^\delta)$ for some $\delta > 0$.

TROPICAL_FOURIER_ANALYSIS (fw_28b4bf95f)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** FOURIER_ANALYTIC
- **Target invariant:** MinimalFourierCoefficient (RealNumber) $\rightarrow$ bounds The discrepancy of a function under tropical Fourier analysis.

Primitives: - TropicalPolynomial (Function): A function defined over a tropical semiring (max-plus or min-plus) with computable coefficients. - TropicalFourierTransform (Transformation): A mapping from tropical polynomials to their Fourier coefficients over a tropical semiring. - DiscrepancyMeasure (Metric): A function quantifying the deviation of a tropical polynomial from uniformity.

Tentative axioms: - A1: TropicalConvolution preserves the tropical semiring structure under composition. - A2: TropicalFourierTransform is invertible when restricted to certain tropical polynomials. - A3: DiscrepancyMeasure is bounded by the maximum absolute value of Fourier coefficients.